Final Year Project Documentation: Property Investment App

Submitted in partial fulfilment of requirements for:
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by

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Declaration of Originality

This is an original work. All references and assistance are acknowledged.

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Date: 05/01/2025

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1. Introduction

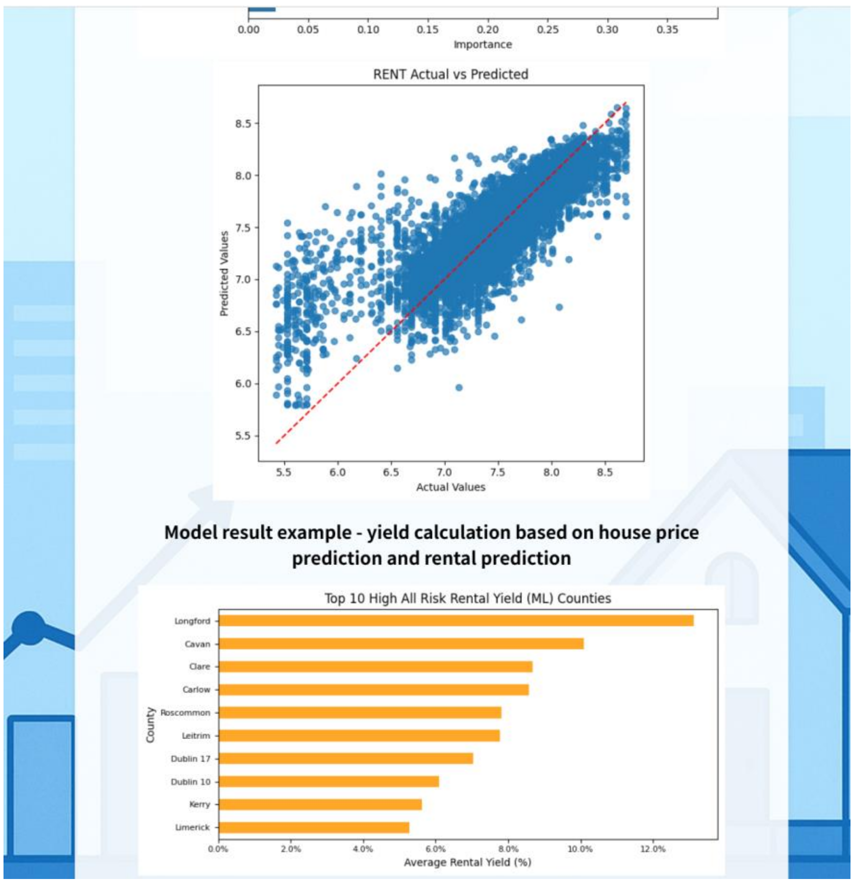
Today, demand for property in Ireland keeps growing. According to the report by Daft.ie, the average price of a property in Ireland increased by 9% last year. Investors wish to get high returns from the property investment, and homebuyers hope to buy a house that is worth the money. A healthy property market can reduce housing vacancy rates and help solve housing problems. However, not every property purchase leads to good outcomes. Bad purchasing property choices can lead to personal financial losses and negatively impact the sustainable development of the community. Making better property purchase decisions can help both achieve personal financial goals and community Development, which links to sustainable development goals (SDG) 8: Decent Work and Economic Growth and SDG 11: Sustainable Cities and Communities.

One of the biggest problems is that many buyers and investors don't have the right tools or information to pick a good property. Prices, rental yield, and long-term value can change a lot depending on different property attributes such as size, number of rooms, and location. It's hard to know what's a good deal, especially for people who are not familiar with the property market. Without clear indicators, it's easy to make bad decisions. And it takes a long time to learn about the market and enter. To help with this, I built a Web-base property investment app that collects and analyzes real-time data from Daft.ie. The app combines machine learning techniques to calculate property yield mainly based on location and number of rooms. The main target users of the app are investors, as the app can help them find stable investments with good returns. It's also very valuable for homebuyers, helping them choose properties that can hold their value over time and select suitable areas to live in.

2. Design

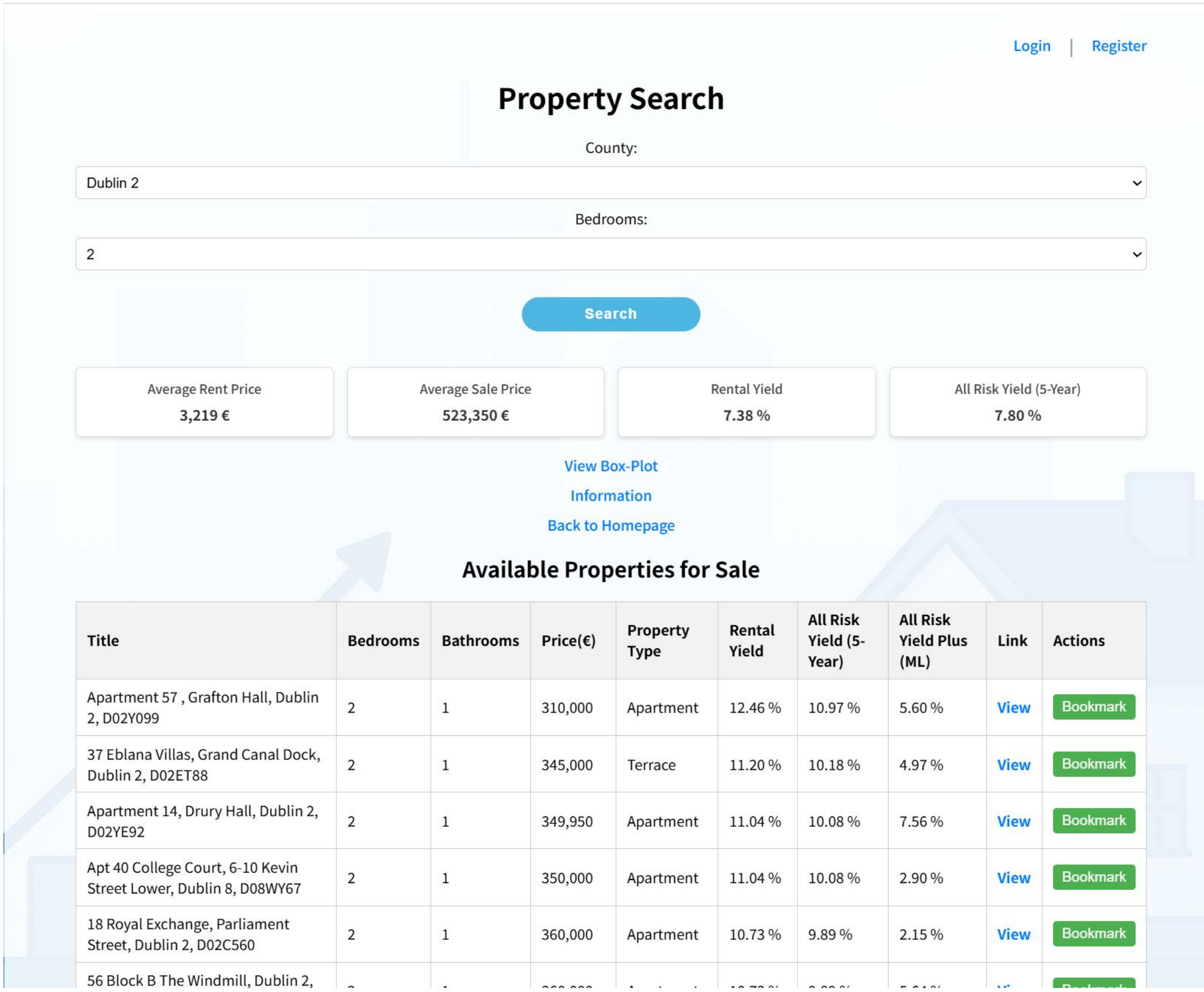
My app has two main functions: market analysis and property search. Market analysis is based on the visualization of existing data. These visualization charts are dynamic, and new charts are generated every time the data is updated. They include the average price trend over the past year, the price distribution frequency of properties, the relationship between property prices and sizes, the price distribution of different types of properties, the counties of cities with the highest and lowest average property prices, counties ranked by simple yield, etc. Example screenshot of this page:

From these diagrams, users can have a basic understanding of the Irish property market. For example, they can learn the market trend from the time series of house prices, and plan their budget from the distribution of house prices. For investors, they can plan the target area for their property investment from the ranking of provinces with the highest yields, and avoid provinces with low yields. For home buyers, they can buy good and cheap properties by observing areas with low average house prices. At the bottom of this page, there is a Machine Learning Insights button. Click it to enter the Machine Learning Insights page, which includes metrics of the machine learning model used in this project, such as R squared, as well as feature importance, and other insights related to the machine learning model. Finally, there is an example output based on machine learning, which is the yield calculated using house prices and rents predicted by machine learning. As shown below:



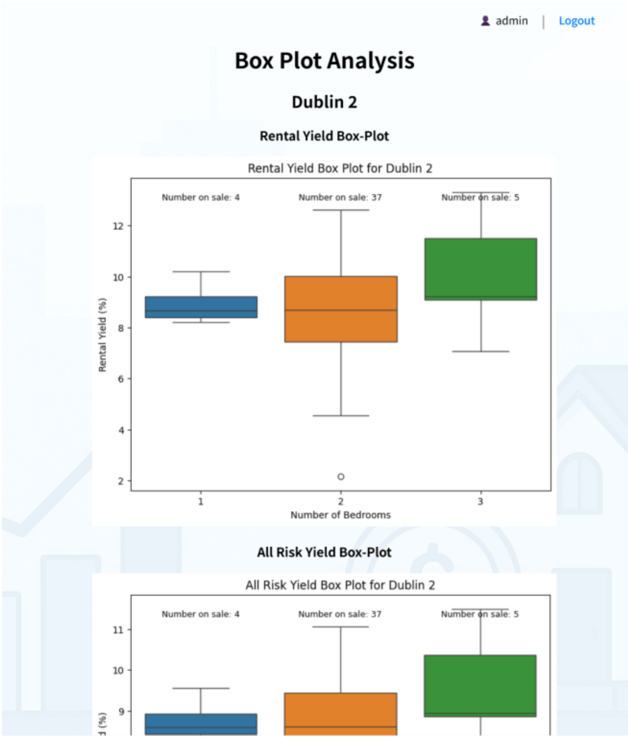
The information on this page is more in-depth and takes into account more factors, including forecasts for house price growth and rent forecasts. This page is mainly for dedicated investors. Because the impact of factors such as area and number of rooms on rent/price in the machine learning model is exactly what investors want to know, which can help them make better investment decisions.

The next important feature is Property Search. In fact, I think this feature is the most important. In this feature, you can select the county you are interested in and the corresponding number of rooms, then click Search, and the corresponding properties for sale will appear. It will also display a lot of indicators, such as the average price and average rent of the number of rooms in the county. At the same time, each property has its own yield analysis, which is divided into Rental Yield, All Risk Yield (5-Year) and All Risk Yield Plus (ML). Rental Yield is a simple yield calculation method, that is, the average annual rent of this type of property/the price of the property. All Risk Yield (5-Year) comes from a replication calculation method I found in some economics papers. Based on IRR, I take into account the investment cycle, review cycle, interest rate, etc., and I set the investment period for this to 5 years. All Risk Yield Plus (ML) is a further upgrade on the basis of All Risk Yield (5-Year), replacing some values calculated using house price/rental growth rate with the prediction results of the machine learning model. Also, You can click on the link behind the corresponding property, and it will jump to the sales page of the property. The following is the screenshot of this page:

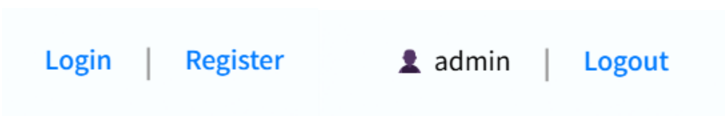


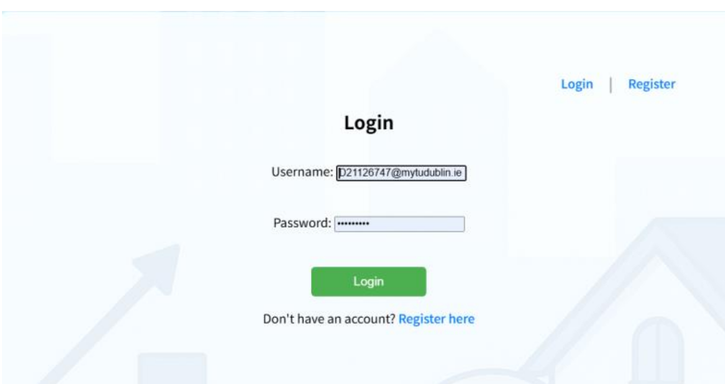
Through these numbers, investors can better weigh the value of these properties and make wise decisions. Specifically, the All Risk Yield Plus (ML) will become more and more reliable as the amount of data increases. Because the machine learning model is trained based on different features, so it can capture the impact of various factors of a property on the price and rent, allowing even people who are not familiar with the property market to make comprehensive decisions and reduce the time cost of researching the market.

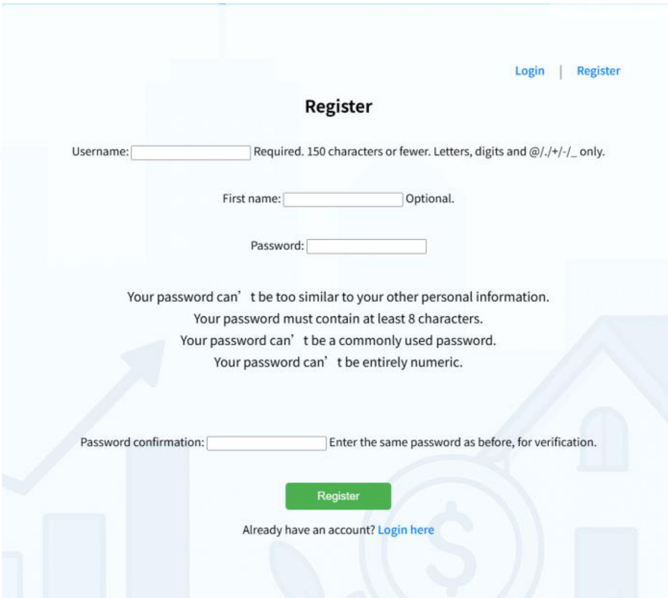
In addition, if you clicking View Box-Plot, you can also view the distribution information of property yields in the selected county:



Apart from these two main functions, I have also designed many other detailed functionalities. For example, user registration and login systems:







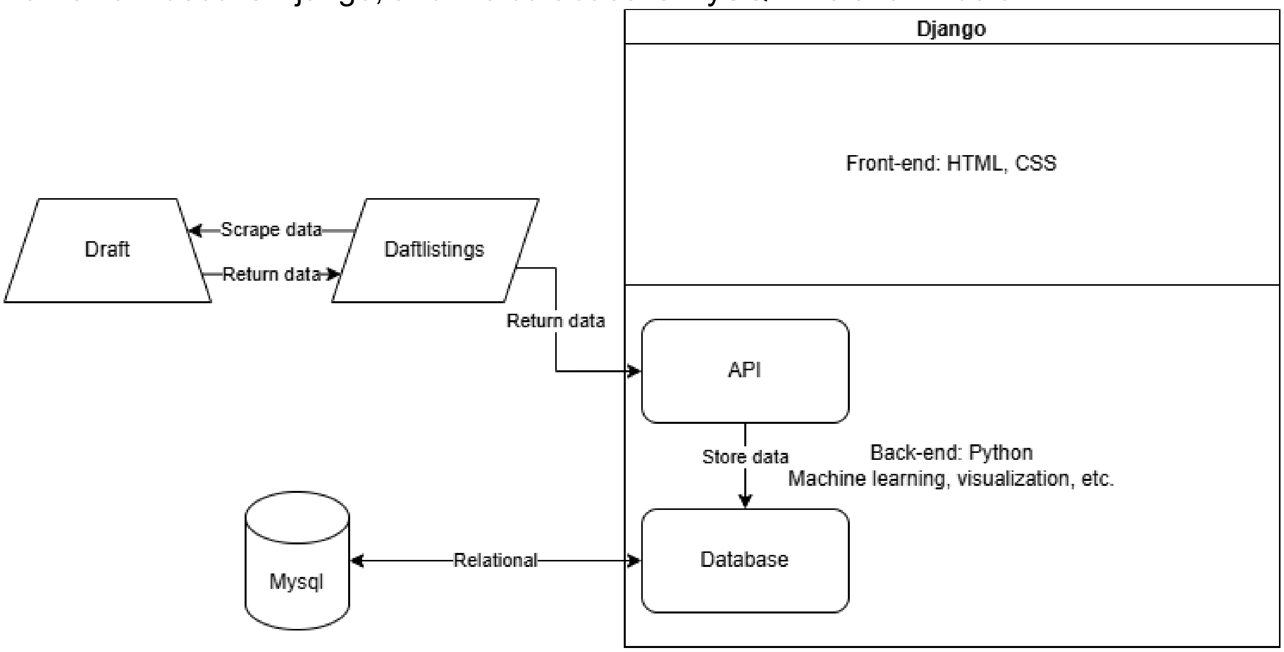
As well as the bookmark feature, which is closely integrated with the login system:

Once logged in, users can mark properties they are interested in. Each user will only see the properties they have marked on the bookmarks page.

My app also includes other features such as a navigation bar that allows users to easily access different pages from anywhere within the application. To avoid getting into too much detail, I'll skip over those smaller features.

3. Technical architecture

The main technologies used in my project are Python, HyperText Markup Language (HTML), Cascading Style Sheets (CSS), and SQL. The framework used is Django, and the database is MySQL. As shown below:



Django is a free and open-source web framework. In this project, HTML and CSS are used for the frontend, handling all UI design and display. Everything the user sees is rendered through them. Python is the programming language used by Django, and all functionalities of the app are implemented with Python. In my app, all interactions, data analysis, and visualizations are also carried out using Python. Furthermore, my App is linked to mySQL, and all the data stored in Django will be stored in mySQL, and through Django I can use Python to interact with it. I also use an API called daftlistings to get data from daft.ie and store it in mysql. Apart from user data, bookmark data, and so on, the main data I use for analysis and visualization is contained in two tables, as shown below:

Propertysale
id : Integer (PK)
title : CharField(255) (nullable)
ecode : CharField(50) (nullable)
Property_Type : CharField(50) (nullable)
price : Decimal(12,2) (nullable)
publish_date : DateTime (nullable)
bedrooms : PositiveInteger (nullable)
bathrooms : PositiveInteger (nullable)
propertySize : CharField(50) (nullable)
county : CharField(100) (nullable)
link : URLField(500) (nullable)
is_active : Boolean (default=True)
rental_yield : Decimal(6,2) (nullable)
all_risk_yield : Decimal(6,2) (nullable)
all_risk_yield_plus : Decimal(6,2) (nullable)

Propertyrent
id : Integer (PK)
title : CharField(255) (nullable)
ecode : CharField(50) (nullable)
Property_Type : CharField(50) (nullable)
price : Decimal(12,2) (nullable)
publish_date : DateTime (nullable)
bedrooms : PositiveInteger (nullable)
bathrooms : PositiveInteger (nullable)
county : CharField(100) (nullable)
link : URLField(500) (nullable)
is_active : Boolean (default=True)

They are registered as models in Django and are automatically created as two tables in MySQL. The two tables are propertysale and propertyrent, which correspond to the records of properties sold and rented. The data from ID to county are used for calculation, analysis and visualization. ecode and county represent different ways of dividing areas; one is postal district and the other is county. The link field stores the URL of the property's listing page on Daft, either for sale or for rent. is_active is used to determine whether the property is available. Also, the price label in both tables serves as the prediction target for my machine learning models.

The remaining three labels in the Properlysale table are the three yields corresponding to each property calculated by the script. They are stored in the table and displayed directly when the app is running instead of being calculated on the spot to reduce loading time. Propertyrent data is used for analysis and rental estimation, but individual rental records are not directly displayed to users. We only show users the properties that are for sale. Since yield represents the potential return if a property is purchased, it is only available for properties that are for sale, not for rental properties.

4. Analysis

As I mentioned earlier, my data source is Daft. Daft covers over 80% of property sales and rental listings in Ireland. I obtained all available for-sale and rental property data from Daft.ie via the API of daftlistings, a library that enables programmatic interaction with Daft.ie. The data is updated daily and stored in a MySQL database connected to Django. And each update will mark whether the property is still available. So far, I have collected data on 16,968 rental properties and 35,838 sales properties. Among them, 1,831 rental properties and 15,028 properties for sale are still available.
Based on these data, I first conducted an exploratory analysis. I exported the properties data for sale and uploaded it to colab. Colab is a hosted Jupyter Notebook service. The programming language used is Python, and you can run the code step by step and output it in cells, which is very suitable for trying before integrating the code into Django.
Because the data type names in SQL are different from those in Python, I first used info() to determine the data size and data structure. Note that because my data is continuously updated, the dataset I analyzed in Colab is different from the one I have now.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 28605 entries, 0 to 28604
Data columns (total 11 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   id             28605 non-null  int64
 1   title          28603 non-null  object
 2   ecode          24604 non-null  object
 3   Property_Type  28596 non-null  object
 4   price          26680 non-null  float64
 5   publish_date   28605 non-null  object
 6   bedrooms       24863 non-null  float64
 7   bathrooms      24431 non-null  float64
 8   propertySize   22290 non-null  object
 9   county         28605 non-null  object
 10  is_active      28605 non-null  int64
dtypes: float64(3), int64(2), object(6)
memory usage: 2.4+ MB
```

Next, I started with data cleaning, I first used describe() to understand the information of the data, such as the maximum and minimum prices. The output made me realize that there were many outliers in the data set:

	id	price	bedrooms	bathrooms	propertySize	is_active
count	2.860500e+04	2.860500e+04	28605.000000	28605.000000	2.860500e+04	28605.000000
mean	5.757929e+06	4.080136e+05	3.222583	2.046810	4.903318e+03	0.488446
std	6.332473e+05	4.687042e+05	1.252376	1.235492	4.860151e+05	0.499875
min	6.646000e+03	1.000000e+04	0.000000	1.000000	0.000000e+00	0.000000
25%	5.794497e+06	2.250000e+05	3.000000	1.000000	9.500000e+01	0.000000
50%	5.913394e+06	3.300000e+05	3.000000	2.000000	1.570000e+02	0.000000
75%	5.994758e+06	4.750000e+05	4.000000	3.000000	3.220000e+02	1.000000
max	6.059871e+06	3.500000e+07	41.000000	25.000000	8.188821e+07	1.000000

Then isnull():

```
Feature-wise null values:
propertySize     6315
bathrooms        4174
ecode            4001
bedrooms         3742
price            1925
Property_Type       9
title               2
id                  0
publish_date        0
county              0
is_active           0
dtype: int64
```

After understanding the missing parts of my data, I filled the null values. Most of the data labels were filled with properties with the same number of rooms in the same county. For small parts such as property categories, bedrooms and bathrooms, I used the most frequent values to fill. During this period, I also converted features with different units such as property size (square meters, acres) to a unified unit. Here are some sample codes:

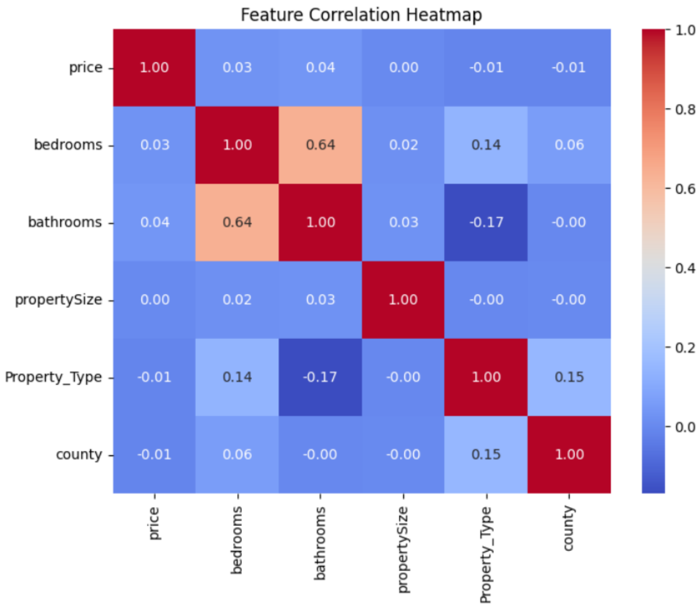
```
#Calculate the average area of the same number of bedrooms
bedroom_grouped_avg_size = df.groupby('bedrooms')['propertySize'].transform('mean')

#Fill rows where propertySize is empty with the average area of the same number of rooms
df['propertySize'].fillna(bedroom_grouped_avg_size, inplace=True)

#If the area of all properties in a bedrooms group is NaN, fill it with the overall median
median_size = df['propertySize'].median()
df['propertySize'].fillna(median_size, inplace=True)
```

```
#Fill bathrooms, bedrooms with the most frequent value
for col in ['bathrooms', 'bedrooms', 'Property_Type']:
    if col in df.columns:
        df[col].fillna(df[col].mode()[0], inplace=True)
```

Next, I made some data visualizations, such as feature correlation heatmaps:



And price distribution digraphs, etc., which is the same as the market analysis part I showed in the Design section. Then, I started trying to do machine learning. I initially tried several models such as RandomForest, XGBoost, combined with data filtering and feature transformation to determine the strategy to maximize accuracy. After determining several combinations that effectively improved accuracy, I performed more complex machine learning. I processed time series and split the time into year, month, and day. I created new features, the time difference between listing and now. Then I tried more machine learning models, such as Linear Regression, Decision Tree, and Gradient Boosting. In the end, the best result was obtained by using the RandomForestRegressor model after applyed Log transformation and log transformation to the data after filtering outliers that were too small and too large. The evaluation metrics are as follow:

```
Model Evaluation Results
MAE: 86871.55
R²: 0.7229
```

Next I applied the same process to the rental property, and slightly different from the sold property, the best model for it was XGboots.
After that, I applied it to the calculation of yield. I calculated three kinds of

return: the first is the ordinary rental yield. That is, the monthly rent multiplied by twelve divided by the house price.
The second is the All Risks Yield, which is the yield after deducting costs and losses. The general All Risks Yield is very simple, that is, the annual yield in the rental yield is changed to pure yield, but the extended formula I used is different from the simple All Risks Yield. I learned this formula by consulting several economics papers, such as Joseph Ataguba's 'A REVIEW OF ALL RISKS YIELD AND IMPLIED RENTAL GROWTH RATE EMBEDDED IN THE EQUATED YIELD HYBRID MODEL OF PROPERTY INVESTMENT VALUATION '. As follow:

$$k_o = e - e\left(\frac{(1+g)^t - 1}{(1+e)^t - 1}\right)$$

In short, the extended All Risks Yield is based on the Equated Yield after taking into account the rental growth rate and the review period, while the Equated Yield is a special internal rate of return (IRR) for real estate investment. For details, please refer to the Property Valuation Principles section in David Isaa's book 'Yields and Interest Rates'. It takes into account operating costs, rent and residual value (property value) after the investment period. After getting the corresponding parameter (growth rate, average cost) through Revenue Irish Tax and Customs, I wrote the code to implement this formula and saved it in the Django management command folder.
The third formula is All Risks Yield plus, which is a formula I improved by myself. I replaced the future rent and future house price calculated by growth rate in Equated Yield with the prediction results of machine learning. The results derived from machine learning combined with various features are more informative and reliable than those calculated using fixed coefficients alone. Combining the code in colab with the code of the previous formula, I converted the complete machine learning training, prediction, and calculation into a python file. And it is also saved in the Django management command folder.

In addition, I also put the data visualization code into the Django management command folder. After that, every time I update the data, I will call these return rate calculation formulas, and calculate them separately for each property data and store them in their tables. New corresponding visualization content is also generated, which is directly displayed when the user views it. It is worth mentioning that with the increase in data volume, the accuracy of machine learning is also gradually improving. The R-squared of the best mechanical model for property price prediction in Colab is 0.7170, and now the latest R-squared in my app has reached 0.9357. The following are the latest evaluation metrics for property price prediction model in the app:

R^2: 0.9357

MAE: €40,626

RMSE: €81,639

5. Conclusions

Overall, my project solves my problem, can provide users with useful information to choose the right property, reduce the learning cost of entering the property market, and help investors make low-risk and high-return investment decisions. It can help advance SDG 8 and SDG 11, particularly in promoting economic opportunity and sustainable urban development. At the same time, my project also has a weakness, that is, the amount of data. Since I did not use a existing dataset and instead collected the data myself, the amount of data available is limited. The more data available, the more accurate the model's predictions become, making my yield analysis more reliable. My idea to improve my project is to cite more comprehensive and more property data. This data is usually enterprise-level. I have tried to contact Daft.ie directly, but I didn't get a response. Therefore, I hope to cooperate with real companies in the future to obtain more valuable data.

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